H

RAFFLES INSTITUTION

2013 Year 6 Preliminary Examination
Higher 2

BIOLOGY — 9648/01

Paper 1 Multiple Choice

30th September 2013

1 hour 15 min

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C**, and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

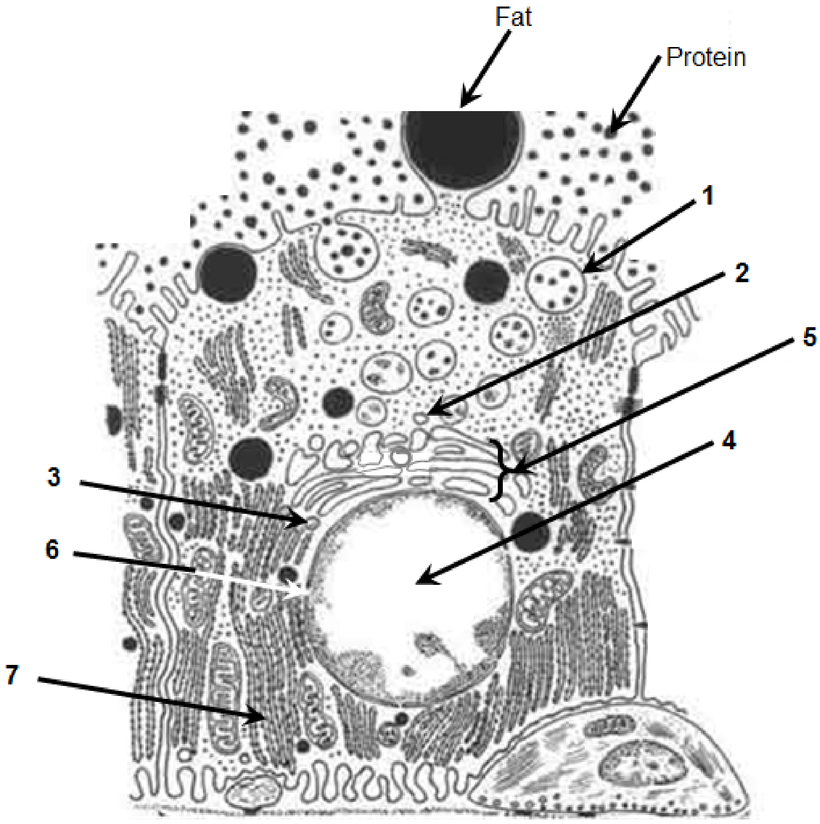
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **22** printed pages.

Raffles Institution
Internal Examination

1. The diagram below shows a lactating mammary secretory cell. In this cell, milk proteins, such as casein, are transcribed and translated, and eventually secreted into the lumen of the mammary gland.



Which of the following shows the most likely sequence of locations involved in this process?

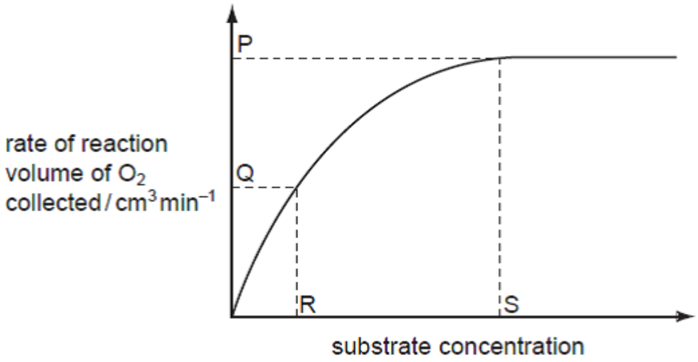
	start						finish
A	6	3	4	7	2	5	1
B	6	4	3	7	5	2	1
C	4	6	7	3	5	2	1
D	4	3	7	6	2	5	1

2. Liver tissue produces an enzyme called catalase which breaks down hydrogen peroxide into water and oxygen.

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time.

Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are **correct**?

1 At **P**, the rate of reaction is limited by the concentration of enzyme.
2 At **Q**, all of the enzyme active sites are occupied by substrate molecules.
3 At **Q**, the rate of reaction is limited by the concentration of the substrate.
4 **R** represents K_m where the reaction rate = V_{max} / 2.
5 At **S**, all of the enzyme active sites are occupied by substrate molecules.

A 1, 3, 4 and 5 only

B 1, 4 and 5 only

C 2 and 3 only

D 2 and 5 only

3. Which two features contribute to the great tensile strength of cellulose?

1 glycosidic bonds linking the long chains of α-1,4-glucose molecules
2 -OH groups of the glucose molecules project outwards and form hydrogen bonds with neighbouring chains
3 strength of the glycosidic bonds between the neighbouring chains of molecules
4 successive glucose molecules are orientated at 180° to each other

A 1 and 3

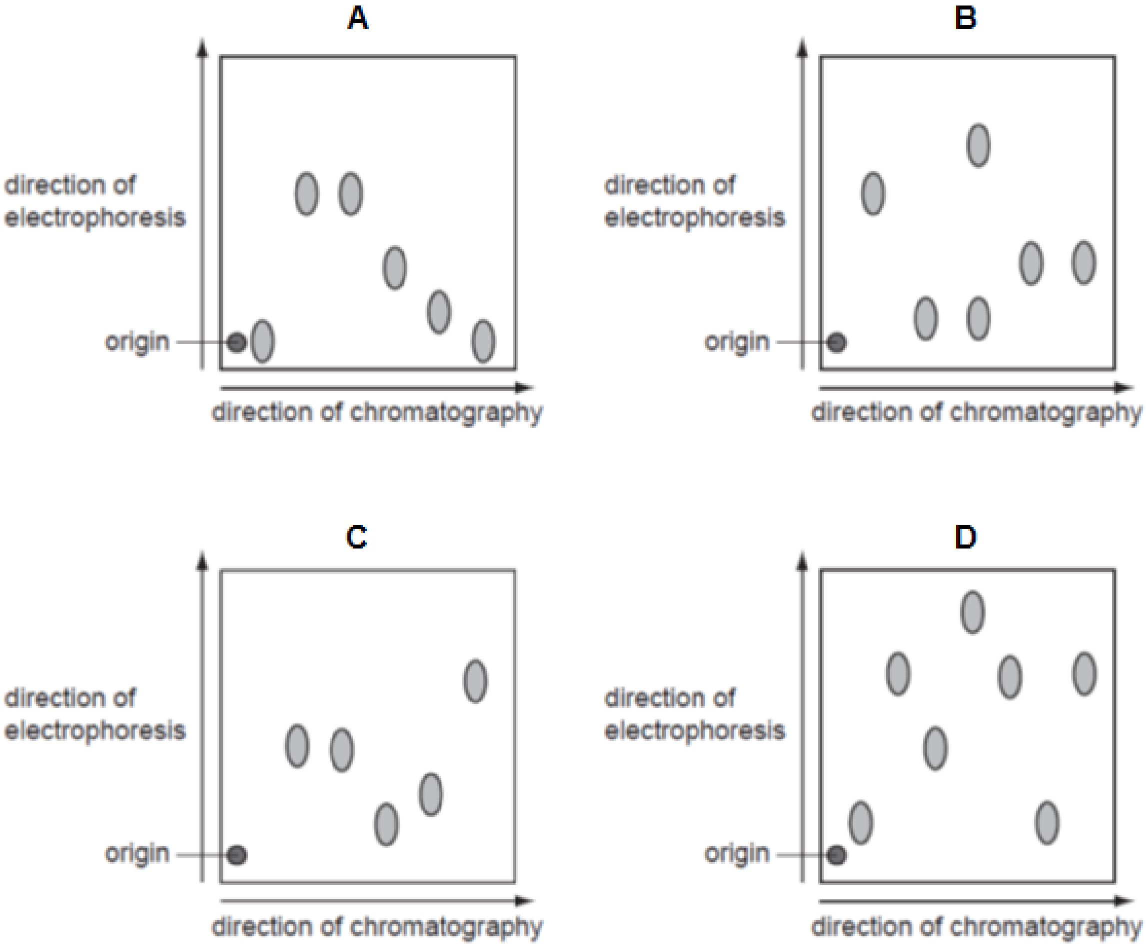
B 1 and 4

C 2 and 3

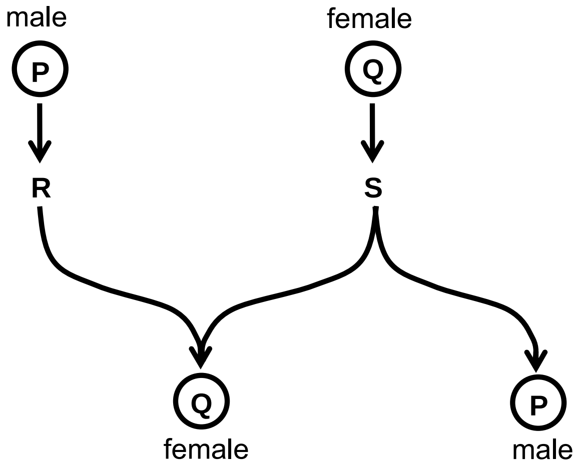
D 2 and 4

4. The diagrams below show the results of an investigation into the composition of different mixtures of amino acids. Each mixture of amino acids was separated using chromatography which separates the amino acids based on their differential solubility in a propanol and aqueous ammonia mixture. Each chromatogram was then turned through 90° and the amino acids separated again by electrophoresis. Electrophoresis separates amino acids based on their charge using an electrical field.

Which diagram shows an amino acid mixture in which the solubility of some of the amino acids is the same but the charge on those particular amino acids is different?



5. Sex determination in some insects such as bees and wasps is not controlled by sex chromosomes.



Using the diagram, which row in the table shows how sex is determined in these insects?

	P	Q	R	S
A	n	n	mitosis	mitosis
B	n	2n	mitosis	meiosis
C	2n	n	meiosis	meiosis
D	2n	2n	meiosis	mitosis

6. Compared to a diploid cell, how much genetic material would the nucleus of a haploid cell of the same organism contain?

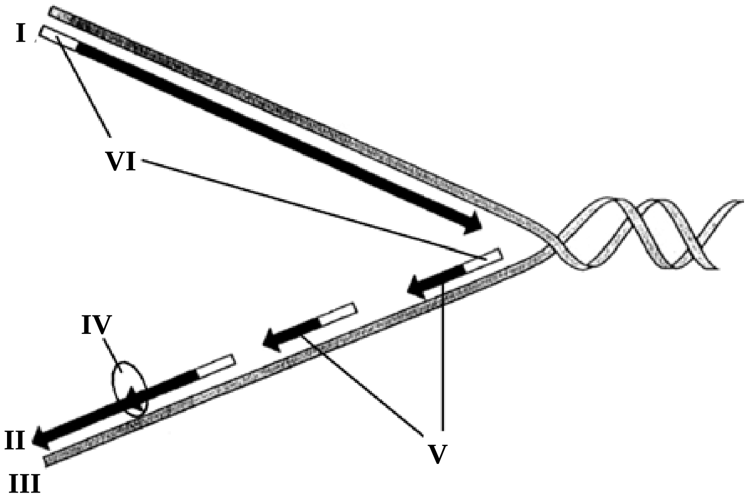
A one quarter

B double

C half

D same amount

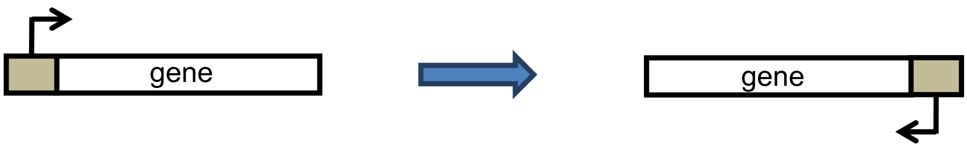
7. The diagram below shows a replication fork.



Which of the following options best describes **I**, **II**, **III**, **IV**, **V** and **VI**?

	I	II	III	IV	V	VI
A	3' end	5' end	lagging strand	DNA polymerase	Okazaki fragments	DNA primer
B	leading strand	lagging strand	5' end	DNA polymerase	Okazaki fragments	RNA primer
C	5' end	3' end	lagging strand	DNA ligase	Okazaki fragments	RNA primer
D	5' end	3' end	template strand	DNA ligase	Okazaki fragments	DNA primer

8. Scientists were able to splice out the promoter of eukaryotic gene and attach it to the end of a gene as shown below. The gene was reintroduced back to the genome of the organism.



Which of the following outcomes is **true**?

A No transcription of the gene takes place as the promoter cannot initiate transcription of the non-template strand.

B Transcription initiates but may prematurely terminate due to a premature STOP codon.

C Transcription takes place but translation cannot take place because the genetic code is no longer the original code.

D Transcription takes place but less translation occurs to obtain functional protein because the mRNA may form a duplex RNA.

9. One of the codons for the amino acid phenylalanine is **UUC**.

Which of the following is the **correct** anticodon of the tRNA carrying phenylalanine?

A 5’ **UUC** 3’

B 5’ **AAG** 3’

C 3’ **AAG** 5’

D 3’ **TTC** 5’

10. Below shows a partial segment of two variant alleles of the gene coding for *Eco*RI in bacteria.

wild type	5’ 3’	Promoter	TTATACCTAATAAATGAACCTGAATTCATGCCGG 3’ AATATGGATTATTTACTTGGACTTAAGTACGGCC 5’
mutant	5’ 3’	Promoter	TTATACCTAATAAATGAACCTGAATTCATGACGG 3’ AATATGGATTATTTACTTGGACTTAAGTACTGCC 5’

Which of the following mutations is seen in this case?

A Nonsense mutation resulting in premature stop codon.

B Silent mutation resulting in no change in amino acid coded.

C Frameshift mutation resulting in change in amino acid sequence downstream of the mutation.

D Missense mutation resulting in a different amino acid coded.

11. A complete reproductive cycle of the T4 bacteriophage requires all of the following processes to occur except

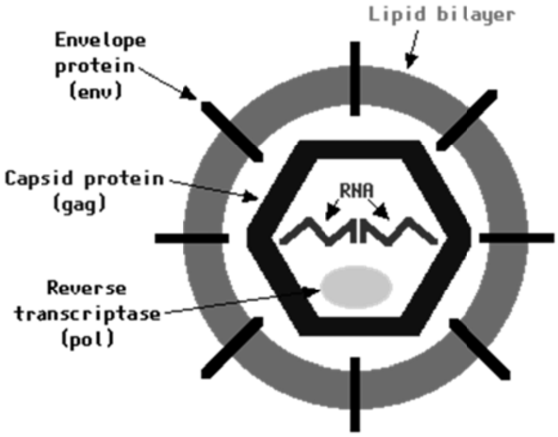
A incorporation of viral DNA into host cell DNA.

B translation of viral mRNA.

C replication of viral genome.

D addition of methyl groups to viral genome.

12. Which of the following statements best distinguishes the type of virus shown below?



A It synthesises DNA from a RNA template.

B It has an envelope that contains some host-cell membrane.

C It is very specific in that it can only infect a single kind of cell.

D It carries two copies of RNA.

13. In the presence of hydrogen sulphide and nitrate ions, the bacterium *Thiomargarita namibiensis* needs to turn on the operons containing the genes for hydrogen sulphide oxidation and genes for nitrate ions reduction. Both operons are turned on in the presence of hydrogen sulphide and nitrate ions. Neither operon is turned on unless hydrogen sulphide and nitrate ions are present. Hydrogen sulphide and nitrate ions are acting as

A inducers.

B activators.

C co-repressors.

D Repressors.

14. Which of the following conditions is likely to interfere with conjugation in bacteria?

A Treatment of the recipient cells with DNase.

B Treatment of the donor cells with strong vibration.

C Treatment of the mating cell pairs with DNase.

D Treatment of the mating cell pairs with strong vibration.

15. Which of the following options is **true** with regards to gene expression in yeast and *E.coli*?

	yeast			*E.coli*		
	polyribosome	polycistronic mRNA	transcription & translation occur simultaneously	polyribosome	polycistronic mRNA	transcription & translation occur simultaneously
A	✗	✓	✓	✗	✗	✗
B	✓	✗	✗	✓	✓	✓
C	✗	✓	✓	✗	✗	✓
D	✓	✗	✗	✗	✓	✓

16. Which of the following statements about transcription in eukaryotes is/are **incorrect**?

1 Acetylation of histones leads to decondensation of heterochromatin to euchromatin for transcription to take place.
2 RNA polymerase, general transcription factors and specific transcription factors make up the transcription initiation complex.
3 Specific transcription factors such as repressors bind to silencers to prevent assembly of the transcription initiation complex.
4 The binding of enhancer proteins to activators result in enhanced rate of transcription.

A 1 and 2 only

B 1 and 3 only

C 2 and 4 only

D None of the above

17. The process of cancer involves

A increasing genetic stability within cells.

B Loss-of-function mutation in a proto-ongene such as the *Ras* gene.

C arrest of cell cycle by tumour suppressor gene products.

D angiogenesis which is important for growth and metastasis of aberrant cells.

18. Eye colour of *Drosophila* is determined by the synthesis of two pigments, red and brown. Wild-type flies have brick-red eyes. There are also flies with scarlet (bright-red), brown and white eyes. Flies homozygous for allele ***s'*** have scarlet eyes whereas flies homozygous for allele ***b*** have brown eyes. The double homozygous recessive genotype results in white eyes.

To investigate the inheritance of eye colour in flies, a geneticist carried out 2 crosses as described below.

CROSS 1
Two strains of wild-type flies that are heterozygous for each of 2 genes, ***s'*** and ***b***, affecting eye colour were mated.

Among 240 progeny,

progeny phenotype	number of progeny
wild-type eyes	150
scarlet eyes	36
brown eyes	46
white eyes	8

The results are analysed by chi-squared test to determine if the difference between the observed and expected numbers are significant. The *P*-value obtained is greater than 0.05.

CROSS 2
A test-cross was performed on wild-type flies that are heterozygous for recessive allele of each of 2 genes, ***s'*** and ***b***.

progeny phenotype	percentage / %
wild-type eyes	25.1
scarlet eyes	24.9
brown eyes	25.2
white eyes	24.8

Which statement **correctly** describes the conclusion that can be drawn from the above crosses?

A Cross 1 results give evidence that we should reject the null hypothesis, indicating that the inheritance of the two genes follows Mendelian ratio of 9:3:3:1.

B Cross 1 results give evidence that we should not reject the null hypothesis as there is interaction between the two gene loci.

C Cross 2 results is used to prove that the inheritance of the two genes follows Mendel's law of segregation.

D Cross 2 results are sufficient to prove that the **S/s'** and **B/b** gene loci are not linked.

19. Achondroplastic dwarfism is an autosomal dominant disorder and red-green colour blindness is an X-linked recessive disorder.

An achondroplastic male dwarf with normal vision marries a colour blind woman of normal height. The man's father is 1.7 meters tall while both the woman's parents are of average height.

What is the probability that their son will be colour blind and of normal height?

A 0.25

B 0.5

C 0.75

D 1.0

20. Feathers in poultry can be white or coloured and this is controlled by two genes, P/p' and Q/q. The phenotypes of offspring that are expected from mating two birds, each of which is heterozygous at both loci, are shown in the Punnett square.

gametes	PQ	Pq'	p'Q	p'q
PQ	white feathers	white feathers	white feathers	white feathers
Pq'	white feathers	white feathers	white feathers	white feathers
p'Q	white feathers	white feathers	coloured feathers	coloured feathers
p'q	white feathers	white feathers	coloured feathers	white feathers

Which of the following best explains the proportion of white to coloured feathers in the Punnett square?

A Dominant epistasis in which a suppressor prevents the expression of epistatic gene.

B Dominant epistasis in which the epistatic allele is P.

C Recessive epistasis in which colour is recessive to no colour at one allelic pair.

D Recessive epistasis in which the epistatic allele is p'.

21. *Achillea millefolium*, commonly known as yarrow is a flowering plant that was traditionally used as a herbal medicine to treat wounds and minor bleedings. It is native to temperate regions of the Northern hemisphere.

Phenotypic variation in the species has been extensively studied along an altitudinal gradient from sea level to more than 3000m elevation. It is observed that there is variation in height and biomass. Adaptation to different local environments also results in variation in physiology such as dormancy period and resistance to cold.

Which is the best method to determine if the observed phenotypic variation in the species is due to genotypic variation?

A Searching for chromosomal differences between populations at the extremes of altitude.

B Determine if viable offspring can be produced when plants from phenotypically different populations are crossed.

C Determine if hybrids between phenotypically different populations can grow at altitudes intermediate between the parent populations.

D Determine if phenotypic differences are maintained when plants from different altitudes are grown under one environmental condition.

22. A child with Down syndrome has the genotype P^1 P^2 P^3 for a polymorphism on chromosome 21 that has four different alleles — allele P^1, allele P^2, allele P^3, allele P^4. The child's mother has the genotype P^1 P^2 and the father has the genotype P^3 P^4.

In which parent did non-disjunction occur, and did this event occur in the first or second meiotic division?

A father; Meiosis I

B father; Meiosis II

C mother; Meiosis I

D mother; Meiosis II

23. Rotene and oligomycin are two metabolic poisons which affect cellular respiration. The effects of rotene and oligomycin on aerobic respiration are summarised in the table.

	ability to use glucose	**ability to use oxygen**	**ATP yield**
rotene	yes	no	decreases
oligomycin	yes	yes	decreases

Which of the following correctly identifies the specific functions of these two metabolic poisons?

	rotene	oligomycin
A	dissipate proton gradient	inhibits ATP synthase
B	inhibits ATP synthase	electron transport inhibitor
C	electron transport inhibitor	inhibits ATP synthase
D	inhibits ATP synthase	dissipate proton gradient

24. The blue dye DCPIP can be converted to colourless DCPIP as shown below:

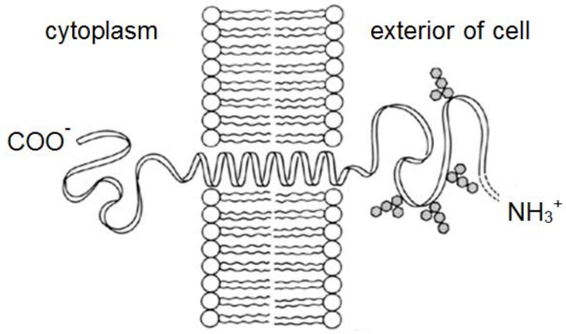
DCPIP (blue) → reduced DCPIP (colourless)

In an experiment, green chloroplast extract was first mixed with DCPIP and the extract turned blue-green. After exposure to 2 hours of light in the presence of ample carbon dioxide and water, the extract became green again.

Which of the following correctly shows the products that were formed after the experiment?

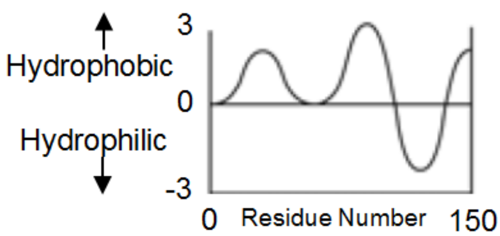
	O_2	ATP	reduced NADP	glucose
A	+	+	–	–
B	–	+	+	+
C	+	–	–	–
D	–	–	+	–

25. Glycophorin, an integral membrane protein, has a single transmembrane α helix.

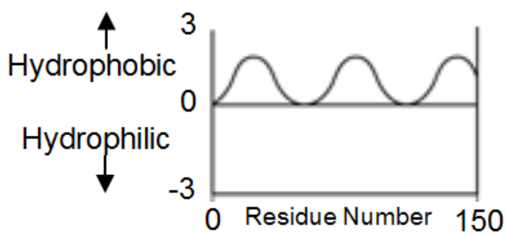


Hydropathy plots are used to estimate the relative hydrophobicity of amino acid residues. Which of the following hydropathy plot most likely represents the transmembrane nature of glycophorin?

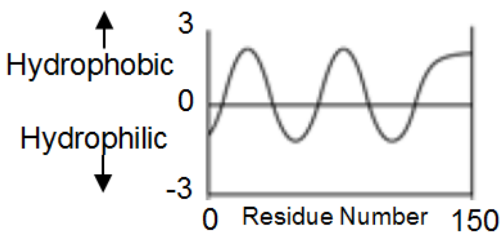
A



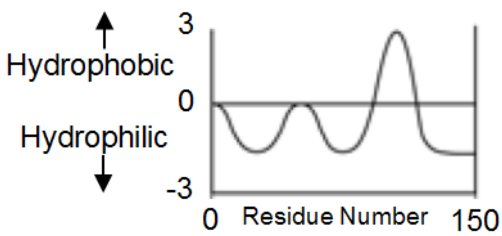
B



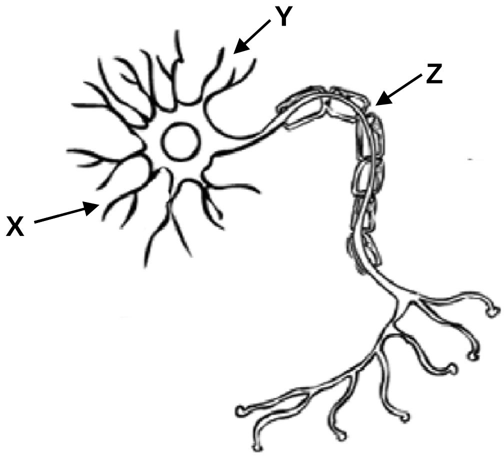
C



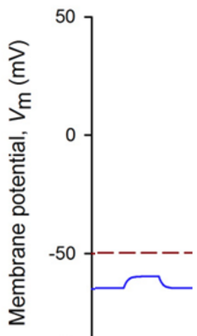
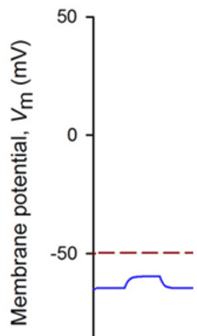
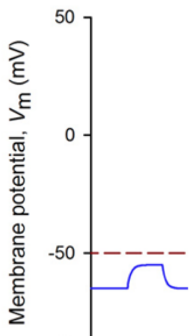
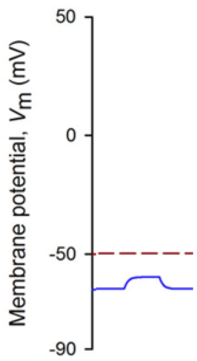
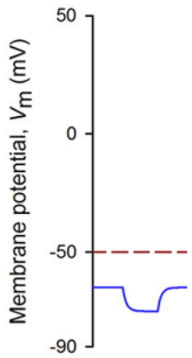
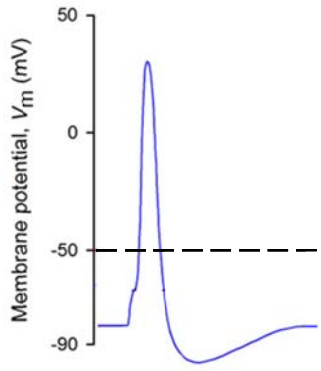
D

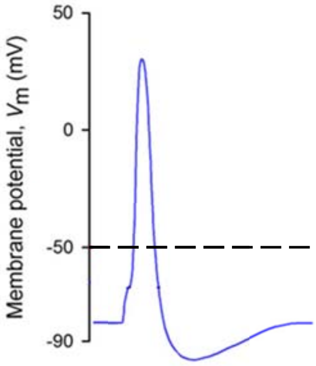
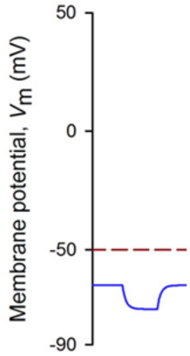
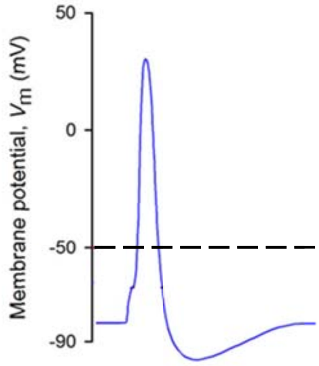
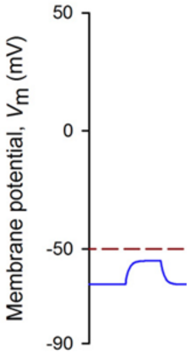
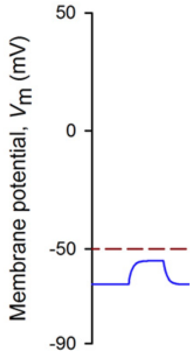
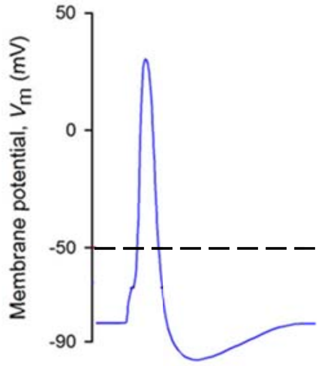


26. The diagram below shows a neuron. Two stimuli are applied, one at **X** and the other at **Y**.



Which of the following **correctly** show the changes in membrane potentials at **X**, **Y** and **Z**?

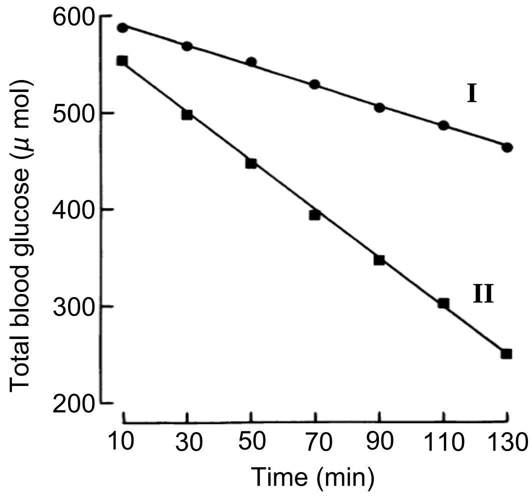
	X	Y	Z
A			
B			

C			
D			

27. It is found that the absence of ATP limits synaptic transmission. Which of the following explains this observation?

	protein affected	effect
A	Na^+-K^+ ATPase	no restoration of Na^+ and K^+ gradient across membrane
B	acetylcholinesterase	acetylcholine cannot be broken down and hence remains in synaptic cleft
C	Ca^{2+} pump	no restoration of Ca^{2+} gradient across membrane
D	voltage-gated Ca^{2+} channel	no influx of Ca^{2+} and fusion of synaptic vesicles with membrane

28. The figure below shows the time course of glucose uptake by the working rat heart in the absence (**I**) of insulin and in the presence (**II**) of insulin. The lower the amount of total blood glucose, the greater is the uptake of glucose by the rat heart.



From the given data, which of the following statements can be concluded to be **correct**?

1 Cells in the rat's heart have receptors for insulin.
2 There is increased uptake of glucose by the rat's heart in the presence of insulin.
3 Release of insulin is stimulated by increase in blood glucose level.
4 The rat is given a carbohydrate-rich meal prior to the experiment.

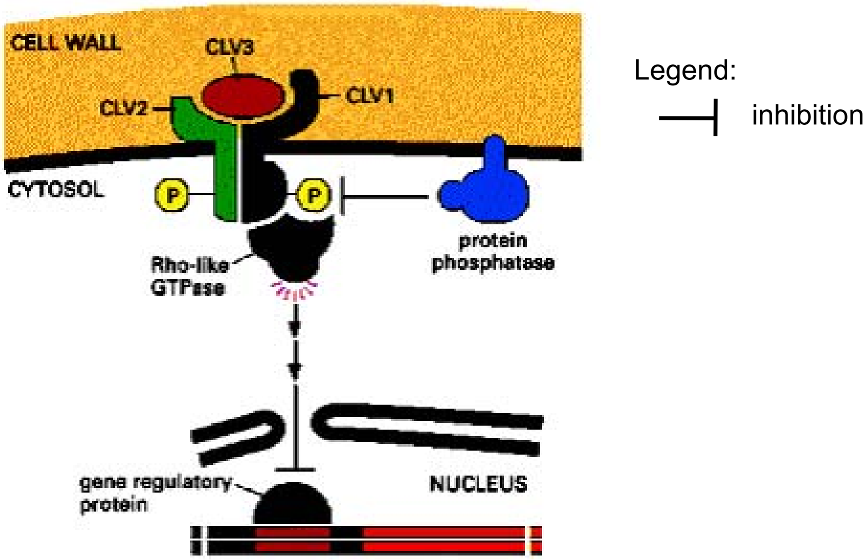
A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 3 and 4 only

29. The diagram below shows the cell signalling pathway involved in cell proliferation and differentiation in the shoot meristem.



Which one of the following sequences regarding the activation of the above signalling pathway is **correct**?

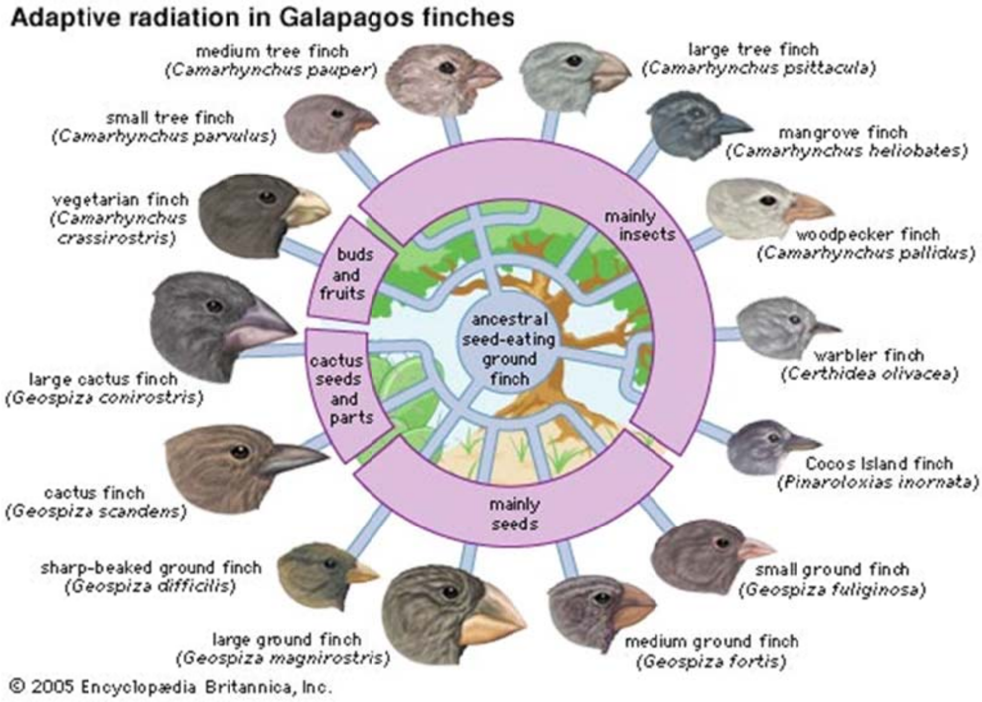
A binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → autophosphorylation of serine/threonine residues → activation of Rho-like protein → inhibition of gene transcription

B dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription

C binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription.

D dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription

30. The Galapagos Islands are a group of volcanic islands in the eastern Pacific Ocean, about 1000km from South America. The diagram below illustrates the finches found on the islands; they resemble each other closely but differ in their feeding habits and in the shape of their beaks.



Which statement below explains how so many distinct and yet similar species arose?

A As a result of the different food sources available, the ancestral species acquired different mutations which resulted in the emergence of a variety of beak shapes.

B The finches were subjected to similar selection pressures which resulted in similar phenotypic characteristics in the distinct populations.

C The difference in food sources served as a selection pressure, allowing only finches with certain beak shapes to survive and reproduce.

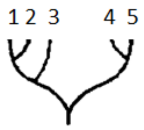
D The finches migrated only to environments that they are already adapted to; finches with different phenotypes migrated to different islands.

31. Five species, **1**, **2**, **3**, **4** and **5** possess the enzyme cytochrome c oxidise. The primary structure of the enzyme was determined for each species and the number of amino acid differences between species is given below.

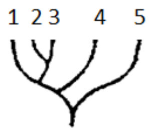
	1	**2**	**3**	**4**	**5**
1	0	7	8	20	23
2	7	0	3	19	22
3	8	3	0	18	21
4	20	19	18	0	20
5	23	22	21	20	0

From the above evidence, which one of the following represents the probable evolutionary relationships between the species?

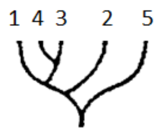
A



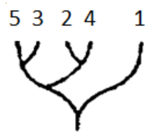
B



C



D



32. An experiment is conducted using a small population of 25 stoneflies with the frequency of allele ***A*** at 50% (i.e. 1 in 2 alleles present in the population is allele ***A***).

The graph below shows the changes in the frequency of allele ***A*** with subsequent generations. The experiment is repeated several times with the same condition and each trial is represented by a line on the graph.

All ***AA***

frequency of allele ***A***

Allele ***A*** lost

generation (25 stoneflies at the start of each)

Which of the following statement(s) is/are **correct**?

1 The populations were undergoing genetic drift.
2 The populations were undergoing selection.
3 Allele ***A*** will be lost in approximately 50% of such trials.
4 Over time, 2 separate species will emerge.

A 1 only

B 2 only

C 1 and 3 only

D 2, 3 and 4 only

33. Which of the following statements is **true**?

	genomic library	cDNA library
A	bacteria plasmid used as vector	bacteriophage used as vector
B	fragments contain introns	no introns
C	used for studying genes responsible for the specialised functions of a particular cell type	used for studying alleles of a particular gene
D	represents a sample of genes expressed at a certain time in the life of an organism	represents a random sample of all the DNA sequences in an organism

34. If the first three nucleotides in a six-nucleotide restriction site are **CTG**, what would the next three nucleotides most likely be?

A AGG

B GTC

C CTG

D CAG

35. The human X-linked disease haemophilia A is a result of a mutation to a gene that codes for Factor VIII, a component of the blood coagulation pathway. The haemophilia allele is recessive to the normal allele.

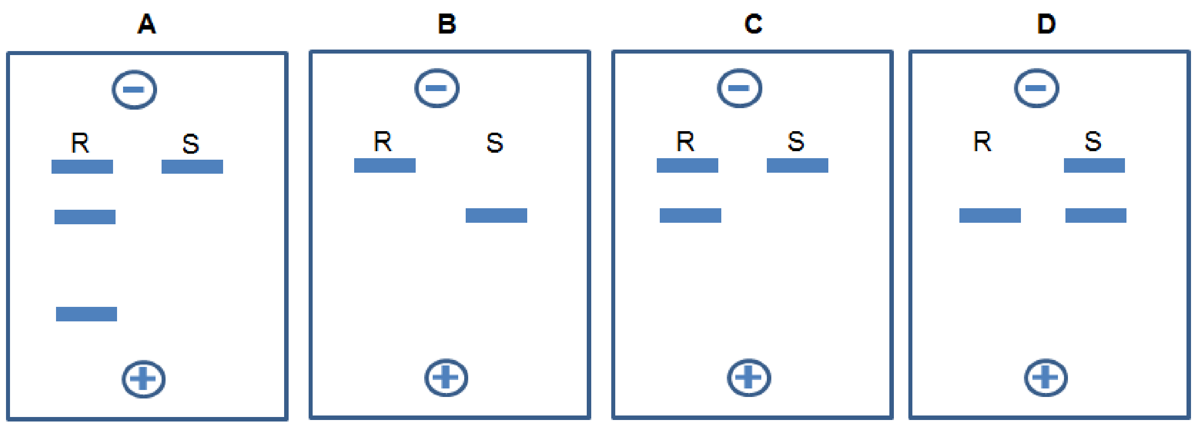
For the gene locus of haemophilia A, the two alleles exist:

normal allele 0.9 kbp 0.3 kbp

probe

hemophilia allele 1.2 kbp

A man who suffers from the disease married a woman who is a carrier of the disease. The couple has 2 children. Child **R** is normal while Child **S** is affected. Which autoradiogram below correctly identifies the children?



36. Which of the following best describes the complete sequence of steps occurring during every cycle of PCR?

1 The primers hybridise to the target DNA.
2 The mixture is heated to a high temperature to denature the double stranded DNA.
3 Fresh *Taq* polymerase is added.
4 *Taq* polymerase extends the primers to make a copy of the target DNA.

A 2, 1, 4

B 3, 2, 4

C 2, 3, 1, 4

D 2, 1, 3, 4

37. Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell. Which statement explains this?

A Genes not required for a particular cell line are methylated.

B mRNA that is not required for a particular cell line is destroyed.

C Genes not required for a particular cell line are removed using restriction enzymes.

D Different stem cells have only the genes required for their particular cell line.

38. Retrovirus is a more suitable vector than liposome for gene therapy as it

A is non-pathogenic and does not trigger an immune response.

B is more specific and causes mutation to the defective allele.

C has a phospholipid bilayer that can fuse with the target cell.

D results in more stable gene integration.

39. Which of the following correctly match the plant tissue culture techniques to their main purposes?

	anther culture	protoplast culture	suspension culture	callus culture
A	to produce hybrid plants	to produce genetically identical plants	to screen for recessive desirable traits	to extract useful secondary metabolites
B	to screen for recessive desirable traits	to produce hybrid plants	to extract useful secondary metabolites	to produce genetically identical plants
C	to produce plantlets which are genetically non-identical	to produce transgenic plants	to trigger shoot and root formation	to produce disease free plants
D	to produce disease free plants	to trigger shoot and root formation	to produce transgenic plants	to produce plantlets which are genetically non-identical

40. *Bt* gene from *Bacillus thuringiensis* may be inserted into cotton plant cells to produce *Bt* cotton plants. Insecticide use and yield in India were compared for *Bt* cotton hybrid (X*Bt*), the same hybrid X but without the *Bt* gene (X_), and another hybrid widely grown in that particular locality (Y). This process was repeated at more than 150 locations. The table below shows the results:

hybrid	**X*Bt***	**X_**	**Y**
mean number of sprays against insects that eat the cotton	0.6	3.7	3.6
mean number of sprays against sap sucking insects	3.6	3.5	3.5
yield / kg ha^{-1}	1500.0	830.0	800.0

The following are the conclusions that are drawn from the data:

1 All insect pests are killed when they consume the *Bt* crop.
2 *Bt* cotton reduces the amount of pesticide used.
3 Both yield and quality of cotton from X*Bt* crop improved.
4 *Bt* cotton increases cost effectiveness.
5 Both X_ and Y hybrids contain susceptible genes to the pest.
6 *Bt* toxin is not found in the plant sap.

Which of the conclusions stated are **correct**?

A 1, 2 and 3

B 2, 3 and 5

C 4, 5 and 6

D 2, 4 and 6

- End of Paper -

Answers:

1.	C	11.	A	21.	D	31.	B
2.	A	12.	A	22.	C	32.	C
3.	D	13.	A	23.	C	33.	B
4.	B	14.	D	24.	A/C	34.	D
5.	B	15.	B	25.	D	35.	C
6.	C	16.	C	26.	D	36.	A
7.	B	17.	D	27.	C	37.	A
8.	D	18.	D	28.	A	38.	D
9.	C	19.	B	29.	A	39.	B
10.	A	20.	B	30.	C	40.	D